

Gitansh Kothari

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Education

University of Toronto, St. George

Bachelor of Applied Science in Computer Engineering

Minors in Artificial Intelligence and Engineering Business.

Toronto, Canada

Sep 2020 – Jun 2025

- GPA: 3.81/4.0
- Graduated with Honors and Dean's List in all semesters.
- Relevant Courses: Neural Networks and Deep Learning, Introduction to AI, Probability, Algorithms and Data Structures, Operating Systems, Matrix Optimizations, Databases, Computer Security.

Work Experience

Magna International

Software Engineering Intern

Toronto, Canada

May 2023 – Jun 2024

- Enhanced drivable area segmentation and object detection pipelines for autonomous driving by training YOLOv5-based models in PyTorch on 45+ classes.
- Converted models to ONNX and TensorRT for efficient inference in C++ on embedded systems with multi-camera real-time input.
- Designed multithreaded preprocessing/post-processing with connected components and temporal tracking for stabilized outputs.
- Refactored system architecture, resolved memory leaks, and integrated CUDA acceleration, reducing processing time by 50%.

Mad Street Den

Machine Learning Intern

Chennai, India

Jun 2022 – Aug 2022

- Implemented and fine-tuned reinforcement learning algorithms (PPO, DQN, A2C) using ViZDoom in dynamic 3D environments.
- Built neural networks for real-time enemy detection and navigation, enhancing agent responsiveness.
- Automated reward shaping and conducted hyperparameter tuning to optimize learning efficiency.

Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Libraries/Frameworks: PyTorch, NumPy, OpenCV, OpenAI Gym, Stable Baselines, C++ STL, React

Tools: Linux, Git, CUDA, cuDNN, ONNX Runtime, TensorRT, Docker, Glade

Projects

Scholarspace AI – Personalized Learning Platform

TypeScript, React, PostgreSQL

Aug 2024 – Apr 2025

- Built an AI-driven platform for classroom management and coding practice with real-time assistance.
- Developed an AI-assisted code editor for guided feedback to students.
- Streamlined classroom creation and assignment workflows for teachers.

Object Detection and Segmentation

Python, PyTorch, OpenCV, NumPy

Feb 2024 – Apr 2024

- Re-architected YOLOv5 for multi-task learning with object detection and semantic segmentation in a single forward pass.
- Modified full training pipeline: model architecture, data loaders, loss functions, and inference code.
- Improved efficiency by avoiding separate models while maintaining strong performance.

Optical Character Recognition

Python, PyTorch, OpenCV

Sep 2022 – Dec 2022

- Implemented a Convolutional Recurrent Neural Network achieving over 80% text recognition accuracy.
- Built a robust training pipeline with data preprocessing, augmentation, and custom loss functions.

UniMaps – GIS Application

C++ STL, Glade

Jan 2022 – Apr 2022

- Developed a GIS-based navigation app with location finding, bookmarking, search, and route planning.
- Implemented the A* pathfinding algorithm and optimized search with a Trie for fast route calculations.

Languages

English (Fluent), Hindi (Fluent), French (Beginner)